

HackVerse 2026 Project Report

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Problem Statement : Blockchain Based Academic Certificate & Digital Verification System.

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Detailed Note about the problem in Real World :-

As competition for jobs increases, candidates may be tempted by fake academic certificates and credentials. Employers then face difficulty determining whether a certificate is genuine, especially when verification is manual, slow, fragmented, or dependent on contacting the issuing institution.

How the problem develops

1. Large number of graduates

Every year, a very large number of students graduate from universities and colleges.

2. Limited employment opportunities

Only a portion of graduates immediately secure jobs. This creates intense competition, particularly for entry-level positions.

3. Pressure on candidates

Some candidates may encounter fraudulent agents or websites offering fake degrees, marksheets, experience certificates, skill certificates, etc.

4. Fake certificates enter recruitment

A candidate can potentially submit a forged PDF/image/document during recruitment.

5. Employer verification becomes difficult

An HR team may have to manually contact the university, check records, send emails, or verify documents through different portals.

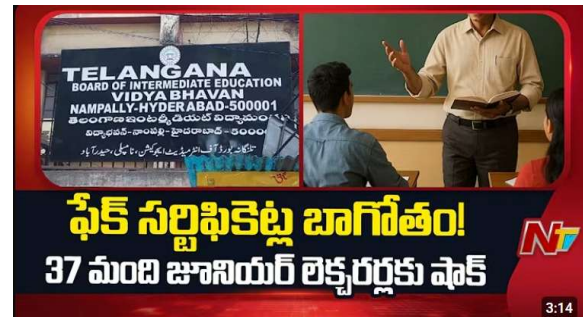
6. Fraud can go undetected

If verification is weak, a person with a fake credential could potentially enter an organization.

7. Society is affected

This damages:

- genuine candidates who earned their qualifications,
- employers,
- educational institutions.



“The issue of fake educational credentials has also entered popular culture, including films and web content that portray fake degrees, forged certificates, and credential fraud. This reflects the fact that credential fraud is not merely a technical issue but a broader social problem.”

Proposed Solution : Credora

Executive Summary & System Overview

Credora is a decentralized credential issuance, storage, and verification platform engineered to eliminate credential fraud and streamline degree verification. The proposed system replaces slow, manual verification workflows with an automated, multi-tiered trust framework combining cryptographic hashing, dual-ledger persistence (an internal SHA-256 blockchain ledger and an EVM smart contract), and a decentralized verifier consensus mechanism.

Core Architectural Pillars

- **Cryptographic Layer:** Utilizes SHA-256 hashing to generate unique digital fingerprints of certificate metadata and documents, embedding quick-response verification codes into generated credentials.
- **Dual-Ledger Persistence Layer:** Combines a local database-backed SHA-256 block ledger engine with public EVM smart contracts deployed via Hardhat on networks such as Polygon Amoy or Sepolia.
- **Decentralized Consensus Layer:** Implements a dual-verifier voting pool where authorized independent reviewers inspect and cast consensus votes before final credential validation.

- **Auditability & Tamper Resistance:** Maintains immutable audit trails of all issuance and revocation actions, paired with interactive sandbox validation tools that simulate document tampering and prove hash invalidation.

User Roles & Access Control Hierarchy

- **Institution:** Authorized entity capable of minting, issuing, batch-uploading, and revoking academic credentials, with public key infrastructure tied to their profile.
- **Student / Recipient:** End-user who can access their credential portfolio, download official PDFs, view embedded verification codes, and share public verification links.
- **Verifier:** Independent third-party reviewer with dedicated review queues to inspect pending credentials and cast cryptographic consensus votes.
- **Admin:** Platform administrator overseeing platform integrity, rate limiting, and system audit logs.



How the System Works

- **Digital Fingerprinting:** Every certificate gets turned into a unique code called a hash, which acts like a digital fingerprint. A quick-scan code is added directly to the certificate so anyone can look it up instantly.
- **Two-Layer Record Keeping:** When a school or university issues a certificate, the system records it in two places: an internal chained database ledger and a public blockchain smart contract. Because each entry is locked to the one before it, past records cannot be edited or erased.
- **Independent Reviewers:** Before a certificate is fully validated, independent reviewers can inspect the submission and vote on whether it is genuine.
- **Instant Verification:** When an employer uploads the document or scans the code, the system recalculates the fingerprint, checks the blockchain records, and confirms the reviewers' votes in seconds.



Speed and Performance in Plain Terms

- **Document Scanning Speed:** Generating the fingerprint depends only on the file size. Larger PDFs take slightly longer to process, but standard files process almost instantly.
- **Certificate Search Speed:** Looking up an existing certificate using its fingerprint or ID is immediate, as the database searches through indexed keys without scanning the entire system.
- **Adding New Records:** Adding a new certificate to the internal chain takes a single quick step, because the system only needs to connect it to the most recent block.
- **Full Chain Audits:** Checking the entire ledger from start to finish scales directly with the number of blocks, as the system checks each link one by one.
- **Reviewer Voting:** Counting votes is virtually instant because it only needs to tally a few reviewer responses per certificate.

Storage and Memory in Plain Terms

- **Low Computer Memory:** The system reads files in small chunks, meaning it uses very little computer memory while generating fingerprints, regardless of document size.
- **Tiny Fingerprint Size:** Every digital fingerprint is converted into a very short, fixed-length string of text that takes up negligible storage space.
- **Database Growth:** The database grows predictably in direct proportion to how many certificates, user accounts, and activity logs are created.
- **Efficient Blockchain Storage:** On the public blockchain, only the essential details—such as the fingerprint, the issuing institution's address, and the status—are saved, keeping transaction and storage fees minimal.

Security and Anti-Tamper Features

- **Strict User Roles:** Access is divided among institutions, students, reviewers, and administrators so only authorized organizations can issue or cancel certificates.
- **Tamper Detection:** If even a single letter, number, or pixel in a certificate is altered, its digital fingerprint changes completely, causing the verification check to fail immediately.

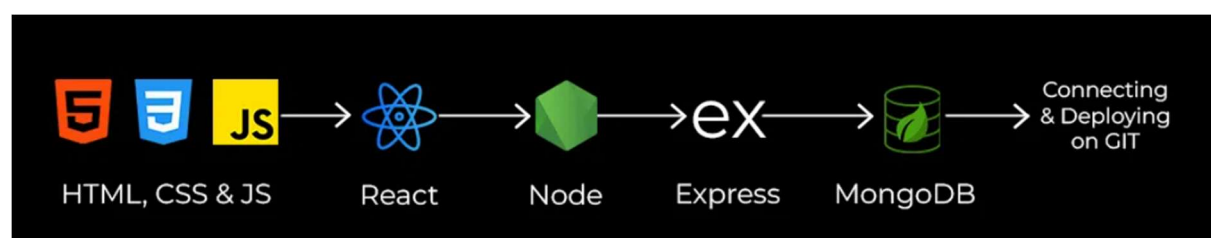
Interactive Sandbox: The platform includes a testing tool that lets users intentionally alter a document to see how the system immediately flags the tampering.

Core Technology Stack

- **Frontend (React.js):** Powers the user interface, role-based dashboards (for institutions, students, and verifiers), real-time verification screens, and interactive tampering sandboxes with clean, responsive styling.
- **Backend (Node.js & Express.js):** Handles the application programming interface (API), manages user authentication and role permissions, generates certificates with dynamic verification codes, and routes verification requests.
- **Database (MongoDB):** Stores structured records including user accounts, institution profiles, certificate metadata, independent reviewer votes, and immutable audit logs.
- **Internal Blockchain Engine (Custom SHA-256 Ledger):** A custom-built, database-backed blockchain engine that links certificate records into sequential cryptographic blocks using SHA-256 hashing, ensuring local records cannot be altered retroactively.
- **Public Blockchain Layer (Solidity & EVM Smart Contracts):** Deploys smart contracts to decentralized networks (such as Polygon Amoy or Sepolia) using tools like Hardhat and Ethers.js to provide an immutable public registry for certificate validation and revocation.

How the MERN Stack and Blockchain Engine Work Together

- **Document Creation & Hashing:** When an institution issues a certificate on the React dashboard, the Express server processes the details, creates a unique digital fingerprint (SHA-256 hash), generates the certificate document with an embedded QR code, and saves the file assets.
- **Dual-Record Storage:** The server writes the certificate details into MongoDB while simultaneously creating a new block in the internal chained ledger and sending a transaction to the public smart contract.
- **Three-Layer Verification:** When a verifier or employer checks a certificate via the React frontend, Express executes three parallel checks: matching the digital fingerprint, checking the chained block history, and querying the public smart contract to confirm validity and consensus approval.



Performance, Speed, and Resource Efficiency in Simple Terms

- **Instant Lookups:** Searching for any certificate by its ID or digital fingerprint is virtually instantaneous because MongoDB uses organized indexes to find records directly without scanning through the entire database.
- **Fast Fingerprinting:** Creating a document fingerprint depends only on the file size. Regular certificate PDFs take a fraction of a second to process and use minimal computer memory.

- **Lightweight Public Blockchain Usage:** Only the essential digital fingerprint and issuer address are written to the public smart contract, avoiding high network transaction fees while keeping the verification speed under a few milliseconds for read operations.
- **Predictable Growth:** Storage grows steadily and linearly as new certificates and audit logs are added, while digital fingerprints always remain tiny, fixed-length strings of text.

Key Security & Fraud Prevention Capabilities

- **Bit-Level Tamper Resistance:** Modifying even a single character, name, or grade in the certificate instantly changes its digital fingerprint, causing automated verification checks to fail immediately.
- **Chained History Protection:** Because each block in the internal ledger points back to the previous block's fingerprint, altering any past entry breaks the entire chain link, which is easily detected by the built-in integrity auditor.
- **Role-Based Access Control:** Strict access boundaries ensure that only verified institutions can issue or revoke credentials, only assigned reviewers can vote on submissions, and students can securely view and share their portfolios.

Scalability

- **Separation of Concerns:** Storing heavy files (PDFs and images) in cloud storage while keeping only compact metadata and digital fingerprints in MongoDB and smart contracts ensures the system scales cleanly without data bloat.
- **Hybrid Ledger Design:** The dual-layer model—handling day-to-day operations and fast verifications through the internal database-backed ledger while using the public blockchain strictly for anchoring and public proof—keeps public network gas fees and network latency minimal.

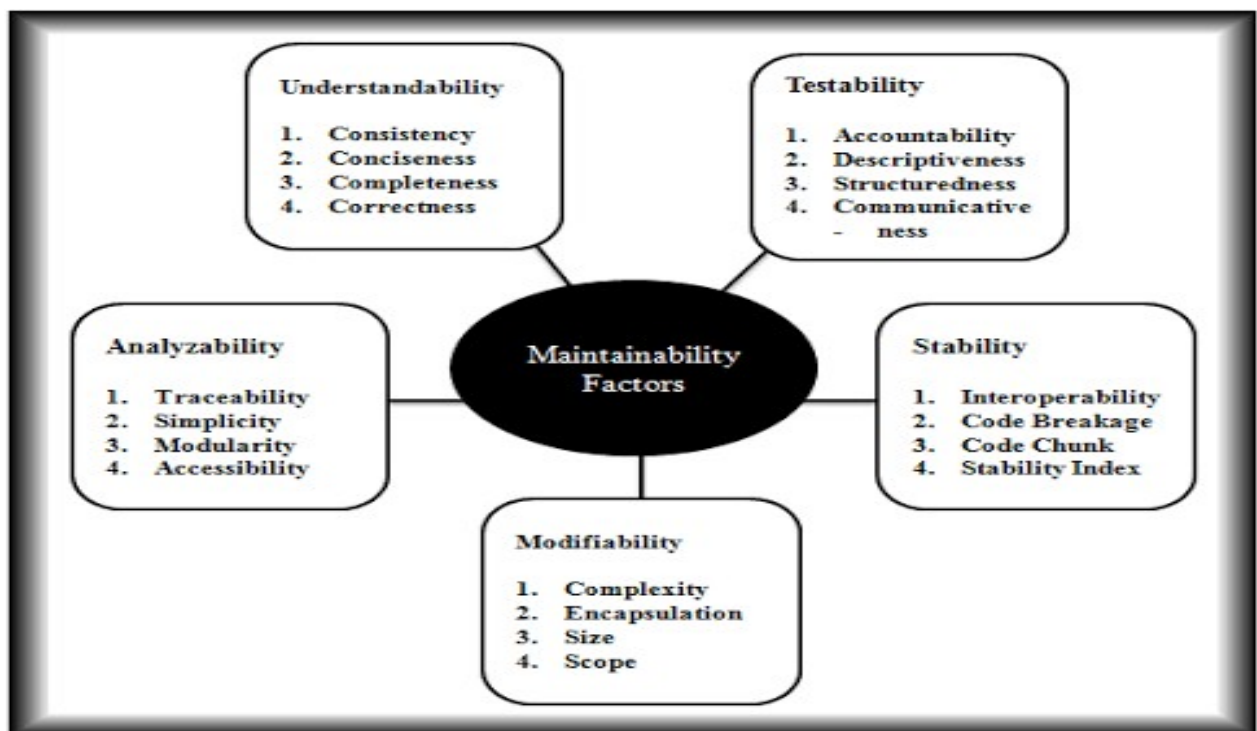


- **Database & Read Throughput:** Direct lookups on indexed fields (like document hashes or certificate IDs) and zero-gas read calls to smart contracts allow the platform to serve thousands of verification checks simultaneously with sub-second response times.
- **Horizontal Expansion:** The stateless Node.js/Express backend and Vite-powered React frontend can easily be scaled across multiple containers or serverless instances behind a load balancer as traffic increases.

Feasibility

- **Accessible, Industry-Standard Stack:** Built entirely on the MERN stack (MongoDB, Express, React, Node.js), enabling rapid deployment, low operational overhead, and straightforward onboarding for full-stack developers.
- **Low-Cost EVM Integration:** Deploying the verification contract to cost-effective Layer-2 or EVM testnets (like Polygon Amoy or Sepolia) avoids high Layer-1 transaction costs while preserving decentralization and tamper resistance.
- **Zero Barrier for Verifiers:** Employers, recruiters, and third-party verifiers do not need cryptocurrency wallets, Web3 plugins, or specialized software; they can verify credentials instantly via standard web browsers or QR code scans.
- **Development & Offline Resilience:** Built-in in-memory database and storage fallbacks allow the entire platform to be demoed, tested, and audited offline or in isolated testing environments without external cloud dependencies.

Maintainability



- **Clean Modular Architecture:** Backend routes, database models, smart contract interfaces, cryptographic services, and UI components are strictly segregated into dedicated directories and layers.
- **Role-Based Access Control (RBAC):** Centralized middleware and route guards keep security, permission checks, and rate limits separate from core business logic, simplifying updates and security patching.
- **Single Source of Contract Truth:** Hardhat tooling and automated scripts simplify smart contract compilation, testing, and multi-network deployments from a unified monorepo.
- **Predictable Data Models:** Well-defined schemas for users, institutions, certificates, consensus votes, and blocks ensure that new features can be added without breaking existing historical records.

Observability

- **Immutable Audit Trail:** Dedicated audit logging tracks every critical system event—including certificate creation, status modifications, revocation requests, and verifier votes—with user and timestamp details.
- **Interactive Blockchain Explorer:** A built-in explorer page allows users and administrators to inspect blocks, verify chain links, examine transaction payloads, and confirm ledger integrity in real time.
- **Live Tamper Detection & Sandbox:** Built-in verification tools and a tamper simulator provide immediate visual feedback if a document hash, database record, or block sequence is modified.
- **Structured Error & Rate Monitoring:** Centralized error-handling and rate-limiting middleware capture failures, protect against abuse, and provide actionable logs for troubleshooting and system monitoring.

Future Scope: Identity and Travel Document Verification

The underlying multi-tier verification architecture can be scaled beyond academic credentials into high-stakes identity and travel document verification:

- **Passport Authenticity:** Government immigration authorities can register unique cryptographic fingerprints of biometric passports on the immutable ledger, enabling foreign border controls and transit agencies to verify document authenticity instantly without exposing sensitive personal records.



- **Decentralized Visa Issuance:** Consulates and embassies can issue tamper-proof digital visas anchored to smart contracts, preventing counterfeit visas, unauthorized sticker alterations, and overstay manipulations through automated status checks.



- **Cross-Border Multi-Party Consensus:** The dual-verifier consensus mechanism can be extended to international immigration checkpoints, allowing multiple authorized border agencies and airline operators to validate travel clearance in real time.
- **Zero-Knowledge Privacy Safeguards:** Incorporating zero-knowledge proofs (ZKPs) in future iterations will allow travelers to prove citizenship, visa validity, or security clearance without revealing full biometric or travel history details to third parties.

Project Conclusion

Credora demonstrates that combining the MERN stack with a hybrid blockchain model successfully addresses the vulnerabilities of traditional credential verification. By pairing an internal chained ledger for fast, cost-effective operations with public smart contracts and verifier consensus, the platform guarantees tamper resistance, operational transparency, and instant verification without requiring complex Web3 setups for end users.

As digital forgery and document manipulation grow more sophisticated, decentralized verification frameworks like Credora provide a robust, scalable blueprint for securing critical credentials—establishing a reliable foundation for the future of academic, professional, and cross-border identity verification.

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